



Emotional Tone After Wins/Losses in Politics and Sports

Timothy Fitch – CSPB 3832 – Final Project



Problem Overview

What Is The Purpose?

Understanding Emotional Responses to Wins and Losses

- How do leaders respond emotionally after positive vs negative outcomes?
- Politics and sports both involve high stakes and high pressure public communication.
- Are the emotional patterns similar or does each profession (domain) express wins/losses differently?
- Useful for sentiment analysis, media studies, and understanding public messaging.



Motivations and Goals

Why Study Wins vs. Losses?

Understanding emotional responses in public communication

- Emotion plays a major role in how leaders communicate after significant events
- Wins often sound positive and losses often sound negative — but how much and in what ways does this vary by domain?

Key Questions:

- Are emotional patterns after wins/losses meaningfully different?
- Do politics and sports show distinct emotional tendencies?
- Can NLP tools capture these differences in a measurable way?



Dataset

Data Gathered:

Political speeches: victory + concession
Sports press conferences: wins + losses
Dataset size: 104 total quotes

Preprocessing steps:

Manual extraction of quotes from source
Cleaned formatting, punctuation, artifacts
Lowercasing + minimal text normalization
Removed filler noise
Labels: win/loss + domain
Saved to structured CSV files

Everything Dabo Swinney said after Clemson's 34-3 loss to No. 1 Georgia

A full transcript of everything Swinney said to reporters postgame.



By AUSTIN HANNON · Aug 31st, 2024, 4:15 PM



John Kerry	loss	11/3/2004	Today, I hope that we can begin the heal	https://www.politics
John Kerry	loss	11/3/2004	It is now clear that there won't be enough	https://www.politics
John Kerry	loss	11/3/2004	All we had was hope and a vision for a bet	https://www.politics
John Kerry	loss	11/3/2004	I thank you from the bottom of my heart.	https://www.politics
John Kerry	loss	11/3/2004	Thanks to Democrats and Republicans ar	https://www.politics
John Kerry	loss	11/3/2004	I will never stop fighting for you.	https://www.politics
John Kerry	loss	11/3/2004	Whether or not our candidates are succe	https://www.politics
John Kerry	loss	11/3/2004	America is in need of unity and longing fo	https://www.politics
John Kerry	loss	11/3/2004	With a grateful heart God bless America.	https://www.politics
Bill Self	loss	3/23/2024	Obviously, we're disappointed in the secc	https://247sports
Bill Self	loss	3/23/2024	I'm proud of our guys. We fought through	https://247sports
Bill Self	loss	3/23/2024	Our program has taken great pride in defe	https://247sports
Bill Self	loss	3/23/2024	Whenever we made mistakes - and we m	https://247sports
Bill Self	loss	3/23/2024	At halftime, I thought we were in decent s	https://247sports
Bill Self	loss	3/23/2024	I'm not going to blame it on energy, but w	https://247sports
Bill Self	loss	3/23/2024	I think for the last month I've been thinki	https://247sports
Bill Self	loss	3/23/2024	We had eight guys on scholarship who we	https://247sports
Bill Self	loss	3/23/2024	Over a season there's a grind When you d	https://247sports

Approach (Methods/Pipeline)

Overall Workflow

- Collect and label quotes (win/loss + domain)
- Clean text and remove filler lines
- Run each quote through a 3-layer emotion pipeline
- Compute Win – Loss differences for each emotion
- Analyze results separately for politics, sports, and combined
- Visualize top differences as bar charts

The pipeline is pretty straightforward consisting of 3 layers:

1. Run VADER for sentiment. (Layer 1)
2. Build EmoLex emotion counts. (Layer 2)
3. Run a GoEmotions RoBERTa model, which uses transformer encoders (Layer 3)
4. Merge everything into one dataset.
5. Compare win vs loss, and then compare politics vs sports.

Everything flows into one CSV.



Tools and Techniques

VADER Sentiment Analyzer (Hutto & Gilbert, 2014)

- Rule-based model for positive/negative valence
- Fast baseline for “overall tone”, performs well on short, expressive statements
- Designed for social media text

NRC EmoLex (Mohammad & Turney, 2013)

- Counts words linked to 8 core emotions
- Useful for broad emotional categories (anger, joy, fear, trust, etc.)

GoEmotions ((Google Research, 2020))

- RoBERTa based transformer trained on 58k reddit comments
- Outputs full probability distribution for fine-grained emotions
- Gives the highest-confidence emotion + intensity

HuggingFace Transformers Library

- Handles model loading, tokenization, and inference
- Makes it easy to combine multiple emotion tools in one pipeline

Final Features Added per Quote

- VADER compound score
- EmoLex category counts
- 28 BERT emotion probabilities
- goem_top_label and goem_top_score (emotion + intensity)



Experiments & Analysis

How the Pipeline Evolved

- Original goal was to classify political vs sports quotes
- Early attempts showed very little difference in surface wording
- Needing deeper emotional features → led to multi-layer analysis

```
Classification report:
              precision    recall  f1-score   support

    politics      1.00      0.18      0.18      10
     sports      0.55      1.00      0.71      11

 accuracy      0.78
 macro avg      0.78      0.55      0.57      21
weighted avg      0.76      0.57      0.46      21
```

```
Confusion matrix (macrotrue, macropred):
```

```
              pred_politics  pred_sports
True politics      1         9
True sports      0        11
```

```
Example predictions:
```

```
Quote: I commit to you that I'm planning to be the governor for the next four years.
```

```
True label: politics
```

```
Predicted label: sports
```

```
Quote: They did fight, they didn't quit, they battled to the end, but at the end of the day we just got our butts kicked.
```

```
True label: sports
```

```
Predicted label: sports
```

```
Quote: I pledge to be a president who seeks not to divide, but to unify... and who will work with all my heart to win the confidence of the whole people.
```

```
True label: politics
```

```
Predicted label: politics
```

```
Quote: This was a test for us... we obviously didn't do very well. But it's the minors, not the final.
```

```
True label: sports
```

```
Predicted label: sports
```

```
Quote: Bruce half, defensively really poor - missed tackles, third-and-long. Their quarterback is really accurate, if we let him hold the ball, he'll find them.
```

```
True label: sports
```

What Each Layer Added to the Experiments

Layer 1 – VADER

- Confirmed basic positive/negative shift after wins/losses
- Showed that surface sentiment changes do exist

Layer 2 – EmoLex

- Revealed which explicit emotions (anger, joy, trust, fear, etc.) were driving changes
- Gave domain (profession) level vocabulary differences

Layer 3 – GoEmotions (Transformer Model)

- Captured subtle emotional patterns not visible in word choice
- Identified fine-grained emotions like admiration, disappointment, gratitude



Results

Politics Only

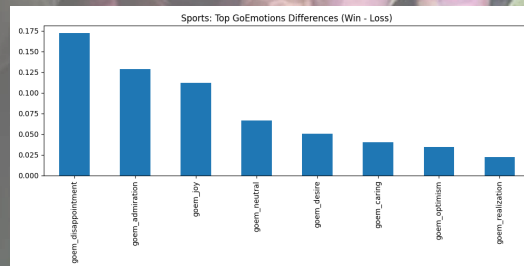
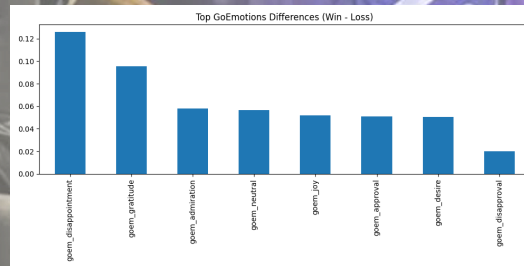
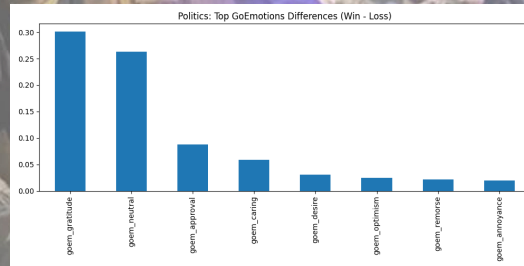
- Wins increase gratitude and neutral tone
- Losses show mild rises in remorse, optimism, and annoyance
- Overall differences are present but small

Sports Only

- Stronger emotional swings than politics
- Losses strongly increase disappointment
- Wins show higher admiration and joy
- Sports figures express emotions more openly

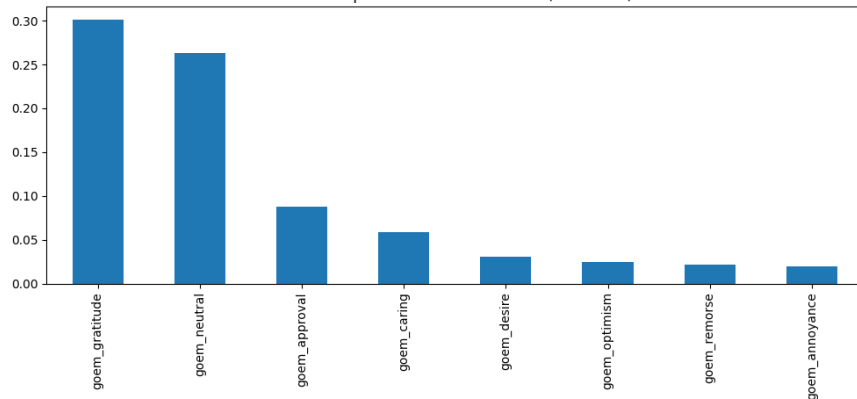
Combined Results (Politics + Sports)

- Clear emotional differences appear when comparing wins vs losses
- Losses show higher disappointment and negative tone
- Wins show stronger gratitude and positive emotions
- Neutral tone also shifts noticeably between outcomes

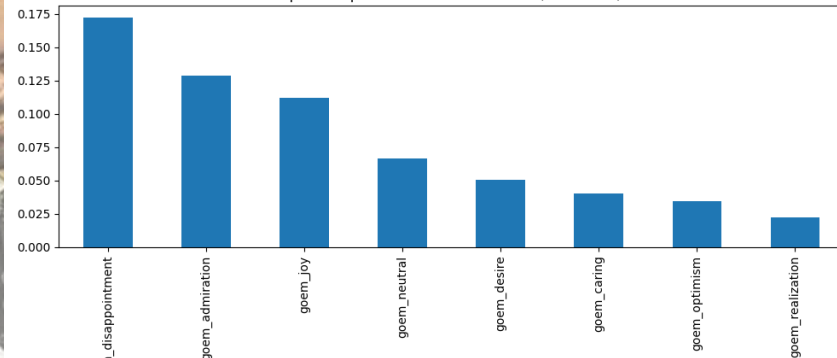


Results

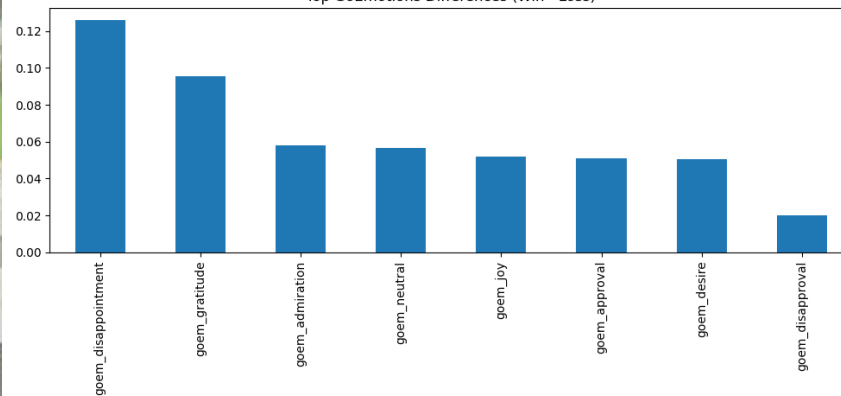
Politics: Top GoEmotions Differences (Win - Loss)



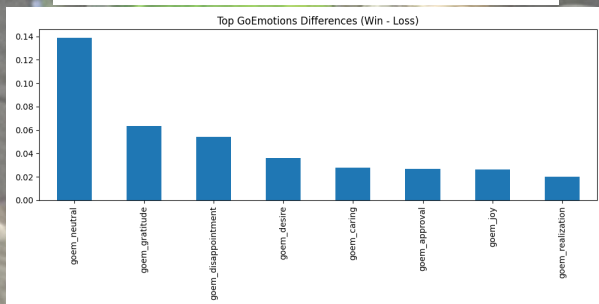
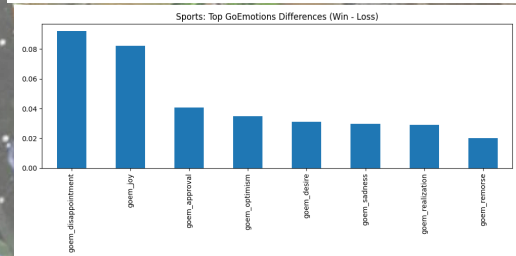
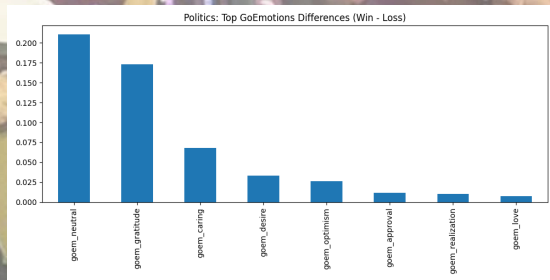
Sports: Top GoEmotions Differences (Win - Loss)



Top GoEmotions Differences (Win - Loss)



Results (Larger Data Set)



To expand and see if the results could be improved upon, I almost doubled the sample size of quotes.

Politics

- Neutral ↑ strongly higher for wins
- Gratitude ↑ higher for wins
- Caring + optimism ↑ moderate win-side increases
- Disappointment ↓ minimal change → politicians remain controlled

Sports

- Disappointment ↑ much higher for losses
- Joy + approval ↑ clearly higher for wins
- Desire, sadness, realization ↑ small swings
- Sports responses show stronger emotional intensity overall

Combined

- Neutral ↑ most increased for wins
- Gratitude ↑ remains a strong win emotion
- Disappointment ↑ still higher for losses but less extreme than Run 1
- Desire, caring, approval, joy ↑ slightly higher for wins

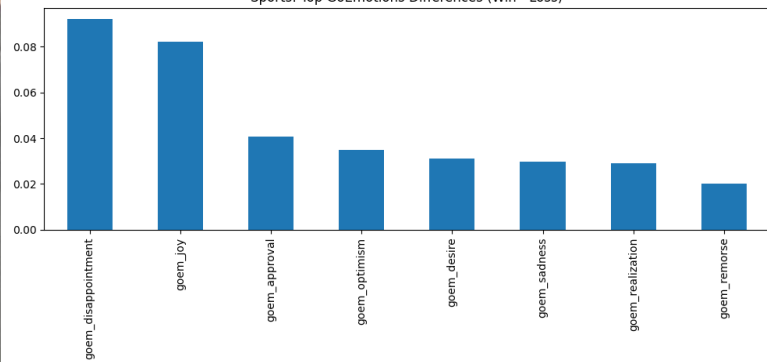
Takeaway (Run 2)

- Sports → big emotional shifts
- Politics → controlled, steady, mostly neutral
- Combined dataset → same overall pattern as Run 1 but smoothed with more data

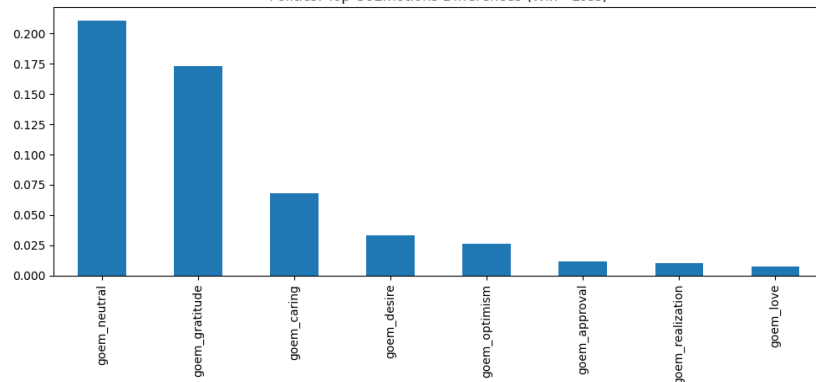


Results (Larger Data Set)

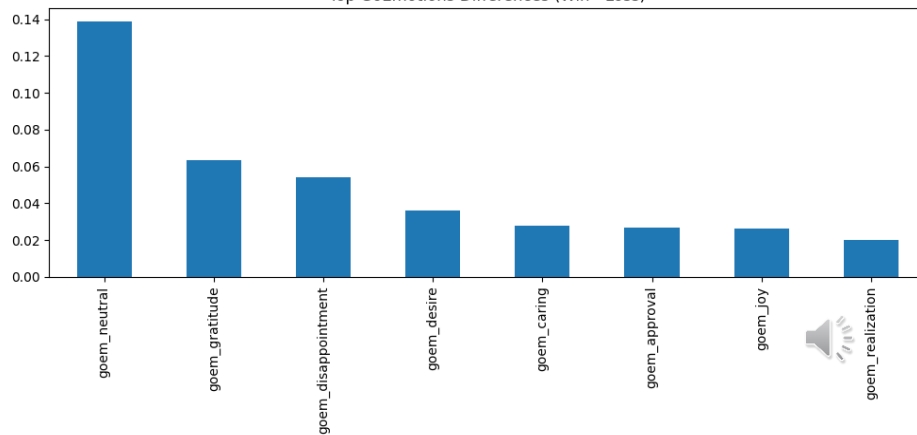
Sports: Top GoEmotions Differences (Win - Loss)



Politics: Top GoEmotions Differences (Win - Loss)



Top GoEmotions Differences (Win - Loss)



Challenges and Lessons Learned

- **Challenges & Lessons Learned**
- Dataset was small and hand-built → results are descriptive rather than statistically strong
- Quote selection + cleaning took time; many lines were neutral or filler
- Early TF-IDF/domain experiments struggled (politics vs sports vocab too similar)
- Transformer-based GoEmotions model captured nuance best
- Larger merged dataset smoothed patterns and confirmed overall trends
- Biggest takeaway: learned full pipeline end-to-end → collecting, labeling, preprocessing, feature extraction, and visualization



Conclusion and Future Work

- Conclusion
- Wins and losses show measurable emotional shifts
 - Wins → gratitude, admiration, joy
 - Losses → disappointment (strongest signal across models)
- Political vs. sports tone differs subtly
 - Politicians: more neutral + gratitude
 - Coaches: stronger disappointment + joy swings
- Patterns consistent across tools (VADER, EmoLex, GoEmotions)
- Larger dataset reinforced earlier trends

Future Work

- Expand dataset for stronger statistical comparisons
- Track emotional trends over time (e.g., election cycles, playoff runs, change over decades)
- Compare underdogs vs. favorites to see if emotional reactions differ
- Add additional domains (business leaders, entertainers, CEOs) for cross-domain emotion comparison
- Try fine-tuning a transformer model specifically on win/loss language

